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Glimmerveen, A.B.; Verhulst, M.M.L.H.; Delnoij, T.; Bonnes, J.L.; Duits, A.A.; Meijer, F.J.A.; Keijzer, H.M.; Hofmeijer, J.

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Clinical paper

Early prediction of long-term cognitive function, emotional distress, and health-related quality of life in cardiac arrest survivors

A.B. Glimmerveen[a,b,*], M.M.L.H. Verhulst[a,b], T. Delnoij[c], J.L. Bonnes[d], H.M. Keijzer[a], J. Hofmeijer[a,b], on behalf of the BROCA investigators[1]

Abstract

Background: Cardiac arrest survivors are at risk of long-term disturbances in cognition, emotional distress, and health-related quality of life (QoL). Early prediction can guide tailored treatment. We evaluated whether early clinical measures predict long-term outcomes.

Methods: In a longitudinal multicenter cohort study, we prospectively included patients who regained consciousness after cardiac arrest. Potential predictors during hospitalization were scores on the Hospital Anxiety and Depression Scale (HADS), Montreal Cognitive Assessment (MoCA), and Barthel Index, and presence of delirium and transient coma >24 h. These were related to cognitive function (MoCA), emotional distress (HADS), and health-related QoL (EQ-5D-5L) at twelve months using multivariate mixed-effects models, with age and sex as covariates. Likelihood ratios were calculated for cognitive impairment (MoCA < 26).

Results: We included 100 patients; 20 % showed cognitive impairment at twelve months and 8 % and 10 % showed signs of anxiety and depression, respectively. Early MoCA showed a borderline significant positive relation with twelve-month MoCA ($\beta = 0.21$, $p = 0.05$). The likelihood ratio for early MoCA < 26 predicting cognitive impairment at twelve months was LR+ 1.29 (95 % CI: 0.40–4.20) and LR- 0.60 (95 % CI: 0.18–1.94). Early HADS was associated with twelve-month HADS ($\beta = 0.47$, 95 % CI: 0.33–0.60, $p < 0.001$), with age negatively associated ($\beta = -0.10$, 95 % CI: -0.18 to -0.02, $p = 0.019$). No other significant relations were found.

Conclusion: MoCA and HADS scores during hospitalization relate to long-term cognitive and emotional outcomes after cardiac arrest. These measures may help identify patients who could benefit from cognitive rehabilitation or psychosocial support.

Keywords: Prediction, Cognitive outcome, Emotional distress, Health-related quality of life, Cardiac arrest survivors, Montreal cognitive assessment

Introduction

Cardiac arrest survivors experience long-term disturbances in cognition and emotional distress.[1–6] These disturbances hinder their participation in daily life.[1,7] International guidelines recommend early screening of patients at risk of long-term cognitive or emotional disturbances, to enable providing targeted treatment. However, there is limited information about the predictive values of potential screening tools.[8]

Early prediction of long-term outcomes after cardiac arrest has been investigated in previous studies. Most studies focused on gross neurological recovery expressed as the cerebral performance category (CPC). A limited number of studies also focused on prediction of cognitive or emotional outcomes. Cross-sectional studies have shown that the Montreal Cognitive Assessment (MoCA) is a valid tool for detecting cognitive impairment in this population, with a MoCA score <26 pointing towards cognitive impairment.[9,10] Longitudinal studies showed that a MoCA score <26 during hospital admission is a potential early predictor of long-term cognitive impairment.[11] Other proposed early predictors associated with long-term cognitive impairment are longer coma duration,[12–14] high S-100B levels,[15,16] elevated neurofilament light chain (NFL) levels,[17] the results of a

* Corresponding author at: P.O. Box 9555, 6800 TA Arnhem, the Netherlands.
E-mail addresses: aglimmerveen@rijnstate.nl, a.b.glimmerveen@utwente.nl (A.B. Glimmerveen).

[1] All collaborators are listed at the end of this paper.

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self-composed bedside neuropsychological test battery,[15,16] and more negative normalized alpha-to-theta ratio or lower peak frequency in the electroencephalogram (EEG).[18] Long-term emotional distress has been associated with younger age,[19,20] longer time till awakening,[20,21] and female sex.[19,22]

Studies on quality of life (QoL) showed divergent results. In some studies, the overall long-term health-related QoL of cardiac arrest survivors was comparable to the general population despite cognitive or emotional impairments.[22–24] Other studies, however, found clear associations between health-related QoL and cognitive complaints, emotional disturbances, and fatigue.[2] Female sex and duration of no flow are probably associated with long-term reductions in some aspects of QoL.[23,25,26]

However, while studies on early screening tests and clinical markers point towards possible associations, none of the proposed markers has been established as a reliable predictor of long term recovery. Further research into early prediction of long-term cognitive, emotional, and QoL outcomes can contribute to optimizing outcome prediction and selection of patients for targeted therapeutic strategies. In this study we investigate the predictive value of early clinical parameters for long-term outcomes in the domains of cognition, emotional distress, and health-related QoL after cardiac arrest. Results will be used as input for future multimodal prediction models.

Methods

This study is part of a multicenter prospective longitudinal cohort study: the Brain Outcome after Cardiac Arrest-prediction (BROCA-prediction) study (Netherlands Trial Register NL9451). The aim of this study is to establish a multimodal prediction model for cognitive, emotional, and participation outcomes of cardiac arrest survivors. Inclusion started in November 2019 and is ongoing. Participating centers are Rijnstate Hospital (Arnhem), Medisch Spectrum Twente (Enschede), Maastricht UMC (Maastricht), Gelderse Vallei (Ede), Radboud UMC (Nijmegen), and Slingeland Hospital (Doetinchem).

This study was approved by the Medical Ethical Committee Arnhem/Nijmegen (NL69767.091.19) and conducted in accordance with the Declaration of Helsinki.

Population

We included adult out-of-hospital cardiac arrest patients who were successfully resuscitated and regained consciousness (Glasgow Coma Scale score > 8). These patients were classically admitted to a cardiology department or cardiac care unit, either directly after resuscitation or after temporary admission to an intensive care unit (ICU).

Exclusion criteria included preexisting brain damage with a Modified Rankin Scale (mRS) score > 2, progressive neurodegenerative disease, a life expectancy of less than three months due to another medical condition, and insufficient knowledge of the Dutch language to complete questionnaires. We obtained written informed consent from all participants or their legal representatives.

Outcome measures

We collected the following outcome measures at one year after the cardiac arrest.

Global cognitive function

Global cognitive function was evaluated using the MoCA by a member of the study team. The MoCA is a 30-point screening tool[27] validated for patients following cardiac arrest.[9] This screening tool assesses the following cognitive domains: abstract reasoning, attention, concentration, executive functioning, language, memory, orientation, and visuospatial abilities. To minimize the risk of misclassifying individuals with limited formal education as cognitively impaired, a compensation of one point is given to those with 12 years or fewer of education. A score below 26 indicates cognitive impairment.[27]

Emotional distress

We measured emotional distress by using the HADS questionnaire. The HADS consists of 14 items, 7 to measure symptoms of anxiety (HADS-A) and 7 to measure symptoms of depression (HADS-D). Per subscale, scores range between 0 and 21. A higher score indicates more symptoms of anxiety or depression, with a value $\geq$8 indicative of a possible disorder.[28] The total score (0–42) can be considered as a measure of general distress, in which a higher score indicates more general distress.[29,30] Though the HADS is not validated in this patient group, it is recommended by the American Heart Association (AHA) and the European Resuscitation Council (ERC).[8,31]

Health-related quality of life

We measured health-related QoL using the European Quality of Life 5-dimension, 5-levels instrument (EQ-5D-5L). This is a standardized measure of self-reported health status and consists of a short questionnaire and a visual analog scale (EQ VAS).[32] The questionnaire covers five domains (mobility, self-care, usual activities, pain/discomfort, and anxiety/depression), with one question per domain. Responses range from 1 (no problems) to 5 (extreme problems), resulting in a 5-digit health profile (e.g. 11111 indicates no problems in any of the domains). These profiles are then converted into a single health index value (EQ-index) using country-specific value sets (tariffs). This generates a predicted value ranging from -0.446 to 1, in which a score below zero represents a health state worse than death and a score of 1 means full health. Mean score in the Dutch general population is 0.869.[33]

These questionnaires are widely accepted tools for evaluation of cognitive functioning, anxiety and depression, and quality of life.

Potential predictors

We collected the following potential predictors according to study protocol within seven weeks after cardiac arrest, during hospital admission:

- Score on the MoCA obtained by a member of the study team.[27]
- Score on the HADS questionnaire.[28]
- The Barthel index (BI) assessed through distant observation by a nurse. BI is a ten-item scale that assesses the level of functional independence of activities in daily living (ADL) such as bathing and grooming.[34] The BI ranges between 0 (complete dependency) and 20 (complete independency).[35]
- Presence of delirium yes or no. Delirium is a clinical syndrome that is characterized by an alteration in attention, consciousness, and cognition, with a reduced ability to focus, sustain, or shift attention. The presence of delirium symptoms was observed by a bedside nurse based on the delirium observation screening (DOS) scale[36] and Diagnostic Criteria Sets-5 (DSM-5) for delirium.[37,38] Since transient delirious symptoms are common after cardiac arrest, a psychiatrist was only consulted in case of severe persistent delirium upon request by the treating cardiologist.[38]

- Transient coma longer than 24 h yes or no. Patients who are successfully resuscitated from cardiac arrest can regain consciousness immediately after return of spontaneous circulation or initially remain comatose for hours, days, or longer. We categorized patients into two groups: those who remained in a coma for at least 24 h (coma group) and those who regained consciousness within 24 h (non-coma group). All patients in the coma group were treated with 24 h targeted temperature management at 36 °C, which could be extended to 72 h in case of fever according to standard clinical care.

Statistical analyses

Demographic information is presented in a descriptive way, as median and interquartile range (IQR) for continuous variables, and number and percentage for categorical variables.

Long-term outcome measures of global cognitive function, emotional distress, and HR-QoL are visualized with violin plots or histograms.

Since two of our long-term outcome measures were related (i.e., HADS-total and EQ-index, please also see supplementary material S1), we used multivariate mixed effect regression models with random intercept. This approach allowed simultaneous analysis of related outcomes and improved the precision of predictor estimates. We treated age and sex as covariates in our models. Multivariate mixed effects regression models are generally robust to missing outcome data under the missing at random (MAR) assumption. This allows us to retain participants even if some outcome values were missing, thereby minimizing potential bias due to incomplete data. Predictor variables with missing values were handled depending on missingness type and extent. Minor missing items in the HADS were imputed based on within-subscale response patterns. For other predictors, including binary variables and the Barthel Index, listwise exclusion was applied where data were not missing at random or imputation was not justifiable. Model assumptions regarding linearity and homoscedasticity were addressed through the flexible structure of the multivariate mixed effects model, which accounts for residual correlations and outcome-specific variance. Given the focus on joint outcome modeling rather than interpretation of individual predictors, no formal diagnostic plots were generated.

To assess the value of early MoCA score <26 as a predictor of long-term cognitive impairment, we calculated the positive (LR+) and negative likelihood ratio (LR−). An LR close to 1 indicates limited predictive value.

Due to the explorative nature of this investigation, p-values <0.05 were considered statistically significant. Statistical analyses were performed using R version 4.3.1.

Results

During the study period, a total of 153 patients were screened for eligibility (see Fig. 1). Of these, 53 were not included in the study. In the current analysis we included 100 cardiac arrest survivors from six different hospitals during the period of November 1st 2019 and June 16th 2022. Eight patients withdrew from the study and five died within a year, leaving eighty-seven patients for this analysis.

Baseline characteristics and potential predictors

Patient characteristics are summarized in Table 1. The majority of our sample was male (85 %) and the median age was 61 years. Median delay to resuscitation was zero minutes and median time to return of spontaneous circulation was ten minutes.

Sixty-six patients (76 %) had initially been admitted to an intensive care unit, of whom fifty-one (59 % of the total population) had been in a comatose state for more than 24 h. Sixteen patients (18 %) experienced a delirious state during hospital stay.

Collection of potential predictors was done at a median 6 days after cardiac arrest. Median MoCA score was 24. Fifty-four patients (62 %) had a MoCA score <26. Median HADS-total score was 7. Twelve patients (14 %) had a HADS-A score ≥8 and nine patients (10 %) had a HADS-D score ≥8, indicating symptoms of anxiety and depression, respectively. Median BI was 19.

Outcome measures

At twelve months, median MoCA score was 27, with twenty patients (23 %) scoring below 26. Median HADS-total score was 5.5 (IQR 2.3–10.8). Eight patients (9 %) had an anxiety score ≥8 and ten patients (11 %) had a depression score ≥8. Median EQ-index was 0.914. The distributions of long-term outcome scores are visualized in Fig. 2. Additional information on the distribution of the EQ-5D-5L subscale scores can be found in the supplementary material S3.

Relation between early potential predictors and long-term outcomes

The MoCA score during hospital stay was positively related to the MoCA score at twelve months at borderline statistical significance ($\beta = 0.21$, $p = 0.05$). The likelihood ratio for a MoCA score <26 during hospital stay predicting a MoCA score <26 at twelve months (LR+) was 1.29 (95 % CI: 0.40–4.20). The likelihood ratio for a MoCA score ≥26 during hospital stay predicting a MoCA score ≥26 at twelve months (LR−) was 0.60 (95 % CI: 0.18–1.94). Age and sex were not related with the MoCA score at twelve months. MoCA score during hospital stay did not relate to HADS-total score and EQ-index at twelve months (Table 2A).

We found a significant positive relationship between the HADS-total score during hospital admission and the HADS-total score at twelve months after cardiac arrest ($\beta = 0.47$, 95 % CI: 0.33–0.60, $p < 0.001$). Age was negatively associated with the HADS-total score at twelve months ($\beta = -0.10$, 95 % CI: −0.18 to −0.02, $p = 0.019$). Sex was not significantly associated with the HADS-total score at twelve months. HADS-total score during hospital stay was not related to MoCA score or EQ-index at twelve months. (Table 2B).

BI, transient comatose state ≥24 h, or presence of delirium did not relate to any of the long-term outcomes (supplementary material S2).

Discussion

In this prospective longitudinal study, we found that the total HADS score collected in the first week after cardiac arrest related to the level of emotional distress at twelve months. To a lesser extent, the MoCA score in the first week related to global cognitive functioning at twelve months. Additionally, higher age was associated with

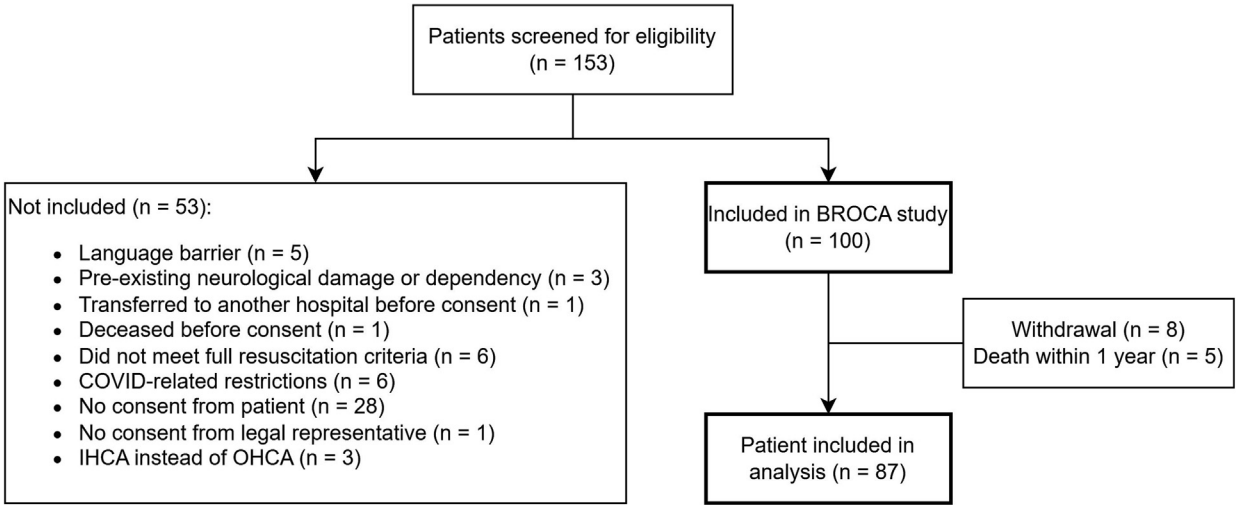

Fig. 1 – Flow of patients through this study during the p eriod of November 1st 2019 and June 16th 2022. IHCA = in-hospital cardiac arrest; OHCA = out-of-hospital cardiac arrest.

less emotional distress at twelve months. This indicates that the MoCA and HADS are probably useful for early screening of cognitive and emotional functioning, to enable tailored treatment. However, likelihood ratios for the prediction of long-term cognitive impairment were relatively low.

During hospitalization, 62 % of our sample showed cognitive impairment as measured with the MoCA. This is consistent with previous studies.[11,39,40] By twelve months, the percentage of patients with cognitive impairment had decreased to 23 %. Previous studies reported that approximately 50 % of cardiac arrest survivors were cognitively impaired at three months,[11,41,42] and 30–40 % at around six months.[43,44] The natural progression of cognitive functioning over time could (partially) explain the difference in prevalence of cognitive impairment between three months in these studies and twelve months in our study. One study identified the MoCA score at four to six days as a significant predictor of cognitive outcome at three months.[11] We found a near significant result for the MoCA score as a predictor of long-term cognitive outcome at twelve months on the group level. Their stronger relationship may be partly explained by the typical trajectory of cognitive recovery over time. Alternatively, the discrepancy could arise from the way we treated the MoCA as a continuous variable in our primary analysis, whereas the other study dichotomized cognition into impaired (MoCA < 26) and non-impaired (MoCA $\geq$ 26). We chose to analyze this variable as a continuous variable, since a score of e.g. 20 is clinically different from 25, but both would be considered impaired. However, our positive likelihood ratio of MoCA < 26 during hospital stay is close to 1, indicating limited predictive value for long-term cognitive impairment at twelve months.

Regarding emotional distress during hospitalization, 14 % and 10 % of the included patients showed symptoms of anxiety and depression respectively, that were likely to be clinically significant. These numbers of patients with symptoms of anxiety and depression during hospitalization are lower than previously described in two systematic reviews and meta-analyses.[45,46] At twelve months, we found similar low levels of signs of anxiety and depression as during hospitalization. This finding of low levels of anxiety and depression is also not in line with the literature.[45–47] However, the proportion of patients identified with symptoms of anxiety and/or depression varies depending on the outcome measure (questionnaire) used. The relation between general distress during hospitalization and general distress at twelve months is in line with previous literature that investigated this relation with outcome at three months.[11] Further, our finding that older age is associated with lower levels of general distress aligns with previous literature.[19] Our findings indicate that early HADS can predict long term emotional distress and help selecting patients that may benefit from treatment to relieve anxiety or depression.

Median level of health-related QoL measured with the EQ-index at twelve months after the cardiac arrest is high in our sample. This is in line with previous studies that reported good QoL at one year after the arrest.[3,24,48,49] If patients report problems, these are mostly in the domains of pain, anxiety, and usual activities, which aligns with a previous study in this population two years after the cardiac arrest.[23] Since a substantial fraction of patients experience cognitive and/or psychological problems and still have a good QoL, the relationship between these two is apparently complex.

We did not find a relation between coma (yes/no) and any of the long-term outcomes. Other studies have shown that longer coma duration was associated with poorer long-term cognitive outcomes.[12,14] Perhaps the duration of coma is important, and our cut-off of 24 h is not clinically relevant. Possibly, coma duration is more relevant than the presence or absence of coma, and our 24-h cut-off is not clinically meaningful. Additionally, in current post-resuscitation care, patients with a Glasgow Coma Scale (GCS) score below 8 are typically sedated for at least 24 h, which means that coma duration in the early phase is largely determined by protocol rather than the patient's neurological status.[8]

Only sixteen percent of our patients had a delirium during hospital stay. This is substantially lower than the previously reported numbers of 33–100 %.[50,51] These higher percentages were mainly found in cohorts of patients after mild therapeutic hypothermia in the ICU. In our sample, 59 % had a coma duration of <24 h, pointing towards less severe postanoxic encephalopathy. In addition, in our patients with a coma duration of >24 h, targeted temperature management

Table 1 – Patient characteristics, disease characteristics, parameters during hospital admission, and outcomes at twelve months.

Characteristic	Value
Sex (male)	74 (85 %)
Age (years)	61 [53.5–69]
Cause cardiac arrest	
Acute coronary syndrome	49 (56 %)
Cardiomyopathy	6 (7 %)
Monogenic arrhythmias	1 (1 %)
Myocarditis	1 (1 %)
Other	19 (22 %)
Unknown	11 (13 %)
Delay till start resuscitation (min)	0 [0–2]
Initial rhythm	
Asystole	1 (1 %)
Pulseless ventricular tachycardia	3 (3 %)
Ventricular fibrillation	76 (77 %)
Unknown	16 (18 %)
Shock administered	
AED	23 (26 %)
Ambulance	34 (39 %)
Both	27 (31 %)
Unknown	3 (3 %)
Time till return of spontaneous circulation(min)	10 [10–15]
ICU stay (yes)	66 (76 %)
ICU stay (days)	2.3 [1.5–4.6]
Ventilation (days)	1 [1–3]
CCU stay (days)	9 [6.5–14.0]
Total stay ICU + CCU	13 [8.5–20]
Potential predictors during hospital admission	
Comatose > 24 h (yes)	51 (59 %)
Timing of assessment potential predictors (days since cardiac arrest)	6 [4–10]
MoCA score	24 [21–26.8]
MoCA < 26	54 (62 %)
HADS-total	7 [3–10]
HADS-A $\geq$ 8	12 (14 %)
HADS-D $\geq$ 8	9 (10 %)
BI	19 [12–20]
Delirium (yes)	16 (18 %)
Outcomes a 12 months	
MoCA score	27 [25–28]
MoCA < 26	20 (23 %)
HADS-total	5.5 [2.3–10.8]
HADS-A $\geq$ 8	8 (9 %)
HADS-D $\geq$ 8	10 (11 %)
EQ-index	0.914 [0.842–1]

All variables are presented as median [IQR] for continuous variables and number (%) for categorical variables. AED = automatic external defibrillator; ICU = intensive care unit; CCU = cardiac care unit; MoCA = Montreal Cognitive Assessment; HADS = Hospital Anxiety and Depression Scale; BI = Barthel Index; EQ-index = European Quality Index.

at 36 degrees was applied instead of hypothermia. These factors may have contributed to less delirium in our cohort. Moreover, the tools used to assess delirium differed: we used an observational checklist, while others combined observation with task-based assessments for the patient. We found no relation between the presence of a delirium during hospital stay and long-term cognitive or emotional outcomes.

Strengths of our study include the multicenter longitudinal design, the standardized collection of clinical parameter data, and the use of a validated cognitive screening tool. There are also some limitations. First, our study may be subject to selection bias, favoring healthier patients. Since patients commit to being followed for a year, this could introduce a selection bias. Second, the relatively small sample size limited the ability to detect moderate associations, even with a conservative modeling strategy. Third, our study lacks external validation. Our current findings will be used as input for a future multimodal validation study. Fourth, we have some missing values at twelve months. However, the multivariate mixed-effects model that

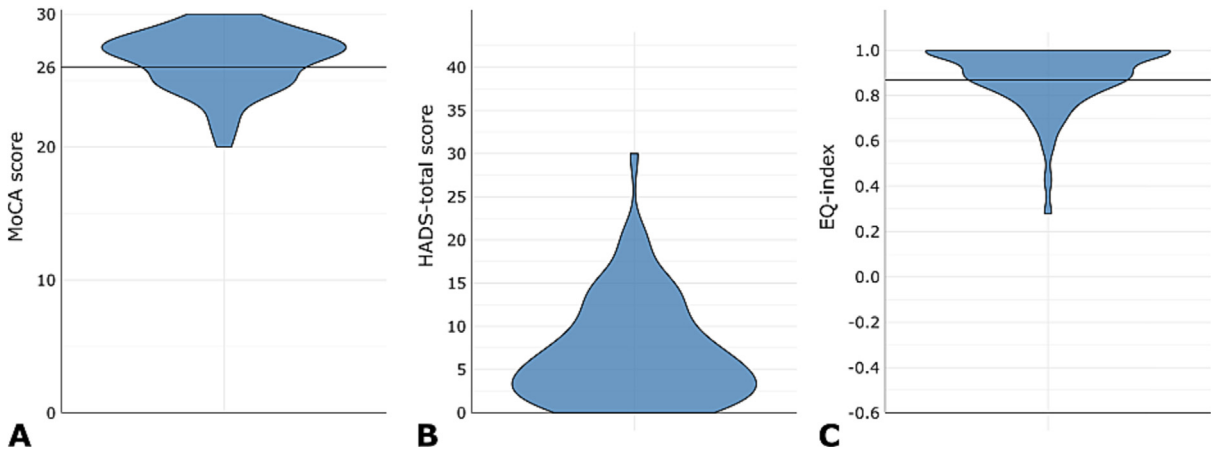

Fig. 2 – Distribution of outcome scores at twelve months, (A) MoCA-score, with the horizontal line depicting a MOCA score of 26, (B) HADS-total score, and (C) EQ-index with the horizontal line at 0.869 indicating the mean score in the Dutch general population.

Table 2 – Results from multivariate mixed effects regression models with random intercept of the relationship between potential early predictors assessed during hospital admission and long-term outcomes at twelve months follow-up. The MoCA during hospitalization is positively related to the MoCA at twelve months. The HADS-total during hospitalization is positively related to the HADS-total at twelve months, while age is negatively related to the HADS-total at twelve months.

2A.	EQ-index 12 months		HADS-total 12 months		MoCA 12 months	
Predictors	**Estimate**	***p*-value**	**Estimate**	***p*-value**	**Estimate**	***p*-value**
Intercept	0.81 [−7.07 to 8.68]	0.84	10.93 [3.06–18.81]	<0.01	24.70 [16.36–33.03]	<0.01
MoCA	0.00 [−0.19 to 0.20]	0.96	−0.00 [−0.20 to 0.20]	0.99	**0.21 [−0.00 to 0.41]**	**0.05**
Age	−0.00 [−0.09 to 0.09]	1.00	−0.07 [−0.16 to 0.02]	0.12	−0.05 [−0.15 to 0.05]	0.36
Sex [female]	−0.10 [−2.52 to 2.32]	0.94	2.07 [−0.35 to 4.49]	0.09	−0.04 [−2.73 to 2.65]	0.98
2B.	**EQ-index 12 months**		**HADS-total 12 months**		**MoCA 12 months**	
Predictors	**Estimate**	***p*-value**	**Estimate**	***p*-value**	**Estimate**	***p*-value**
Intercept	0.97 [−3.96 to 5.89]	0.70	8.96 [4.04–13.89]	<0.01	29.87 [23.97–35.77]	<0.01
HADS-total	−0.00 [−0.13 to 0.13]	0.94	**0.45 [0.32–0.58]**	**<0.01**	−0.00 [−0.14 to 0.14]	0.99
Age	−0.00 [−0.08 to 0.08]	0.99	**−0.09 [−0.17 to −0.01]**	**0.03**	−0.05 [−0.15 to 0.04]	0.26
Sex [female]	−0.09 [−2.31 to 2.13]	0.94	0.89 [−1.33 to 3.12]	0.43	−0.15 [−2.68 to 2.37]	0.91

MoCA = Montreal Cognitive Assessment; HADS = Hospital Anxiety and Depression Scale; EQ-index = European Quality index.

we used can handle missing data when the data are missing at random (MAR), thereby reducing potential bias due to attrition.

Conclusion

Scores on the MoCA and HADS in the first week after cardiac arrest are related to cognitive and emotional function at one year. Although the effect sizes are modest, these early measures can help identify patients who may benefit from targeted rehabilitation treatment.

Ethical statement

The medical Ethical Committee Arnhem/Nijmegen approved the protocol (NL69767.091.19) on 27-05-2019.

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CRediT authorship contribution statement

A.B. Glimmerveen: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **M. M.L.H. Verhulst:** Writing – review & editing, Methodology, Investigation, Formal analysis. **T. Delnoij:** Writing – review & editing, Methodology, Formal analysis. **J.L. Bonnes:** Writing – review & editing, Methodology, Formal analysis. **H.M. Keijzer:** Writing – review & edit-

ing, Supervision, Methodology, Formal analysis. **J. Hofmeijer:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

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Author details

on behalf ofthe BROCA investigators[1] [a]Department of Neurology, Rijnstate Hospital, Arnhem, the Netherlands [b]Clinical Neurophysiology, Technical Medical Centre, University of Twente, Enschede, the Netherlands [c]Department of Intensive Care Medicine, Maastricht University Medical Center, Maastricht, the Netherlands[d]Department of Cardiology, Radboud University Medical Center, Nijmegen, the Netherlands

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